

RISING ELECTRIC VEHICLE REGISTRATIONS AND NEW EU MEASURES: IS EUROPE'S BATTERY INDUSTRY THRIVING?

In 2024, the global electric vehicle (EV) market continued its strong growth, accompanied by the rising demand for lithium-ion batteries. While the Chinese market experienced significant growth, the past year was very challenging for the European market. In the first quarter of 2025, EV registrations in Europe rose again significantly compared to the previous year. At the same time, the EU provided new impetus with measures such as the Clean Industrial Act to strengthen the competitiveness of the battery industry. The analysis of cell and component productions shows that despite difficulties the battery industry in Europe still has great potential to meet the growing demand in the EV market.



2024: Global growth despite stagnation in Europe

With over [17 million EVs sold](#) (battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs)), a new global sales record was achieved in 2024. Compared to the previous year, approximately 3.5 million more electric vehicles were sold worldwide, representing a sales increase of around 25%. However, the EV market developed very differently in the individual regions. While China, supported by [political measures](#), was able to increase sales by over 3 million to 11.6 million electric vehicles, the market in Europe stagnated at around 3.1 million newly registered vehicles.

In the first quarter of 2025, the European electric vehicle market rebounded and resumed its upward trajectory. Year-on-year, sales rose by around 20% to a total of 841,000 BEVs and PHEVs. In addition to electric vehicles that can be externally charged, the number of hybrid electric vehicles¹ (HEV) without external charging capability

Figure 1: Global sales BEV + PHEV, broken down by different regions.

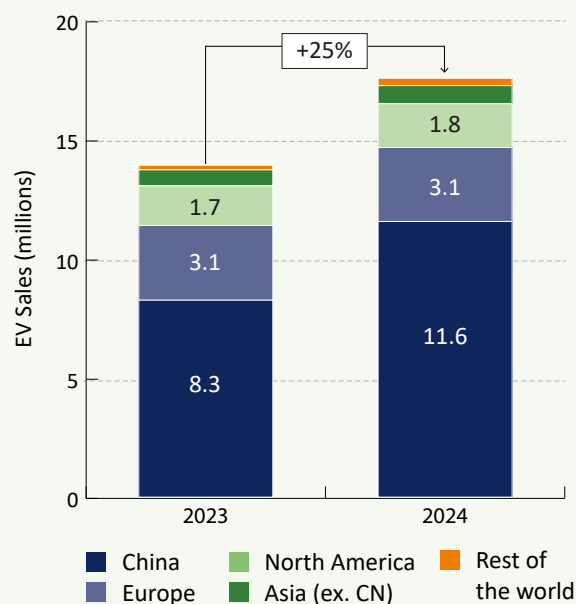
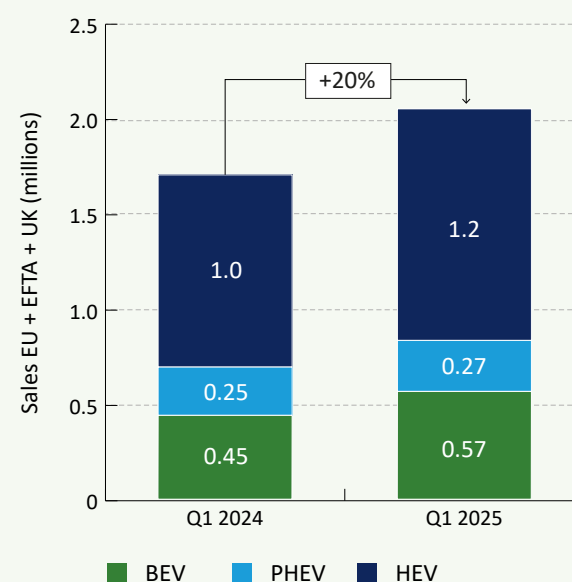


Figure 2: BEV, PHEV and HEV sales in Europe (EU + EFTA + UK) in the first quarter of 2024 and 2025.

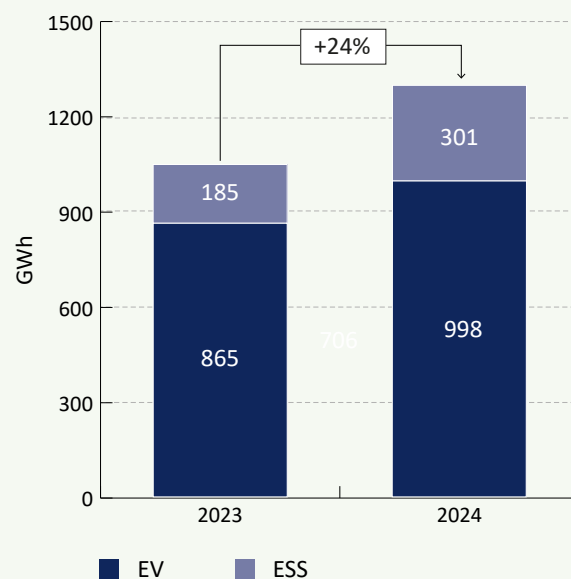


¹ A distinction is made between mild hybrids and full hybrids. The lithium-ion batteries installed in mild hybrids have a battery size of around 0.5 to 1 kWh. In full hybrids, the battery size is around 0.8 to 2 kWh.

is also increasing. Around 1.2 million HEVs were registered in the first quarter, representing a 20% increase compared to the previous year. HEVs typically use lithium-ion batteries, which are significantly [smaller](#) compared to PHEV and BEV batteries, ranging from 0.5 to 2 kWh. However, they nonetheless generate an annual battery demand in the single-digit gigawatt-hour range in Europe. Together, BEVs, PHEVs and HEVs achieved a new registration share of around 60% in the first quarter of 2025, a rise of around 10 percentage points compared to the previous year. The increase in registrations in Europe is partly due to the EU's stricter CO₂ emission requirements. As described in the previous [Market Update](#), car manufacturers are implementing a range of measures to meet emission targets. These include price reductions, a broader variety of model offerings, and an expanded presence in the segment under EUR 25,000. At the beginning of the year, the EU Commission adjusted guidelines for achieving the CO₂ emission targets as part of the [Automotive Action Plan](#). In the period from 2025 to 2027, car manufacturers are allowed to exceed the CO₂ target in one year, provided they compensate the excess in another year. This provides car manufacturer with a bit more flexibility without abandoning the long-term CO₂ emission goals.

The impact of the US tariff policies on the global vehicle market and ultimately on the global EV market is still unclear. According to the recently published [EV Outlook 2025](#) by the IEA, around 10% higher EV sales are expected in the USA compared to the previous year, despite potential tariffs. The termination of the 'Clean Vehicle Tax Credit' discussed as part of the [One Big Beautiful Bill Act](#) could temporarily boost EV sales in the USA, as prospective buyers considering the purchase of an EV might make a quicker buying decision to secure the tax credit. Overall, the *EV Outlook 2025* takes a positive view

Figure 3: Development of the global EV and ESS lithium-ion battery market.

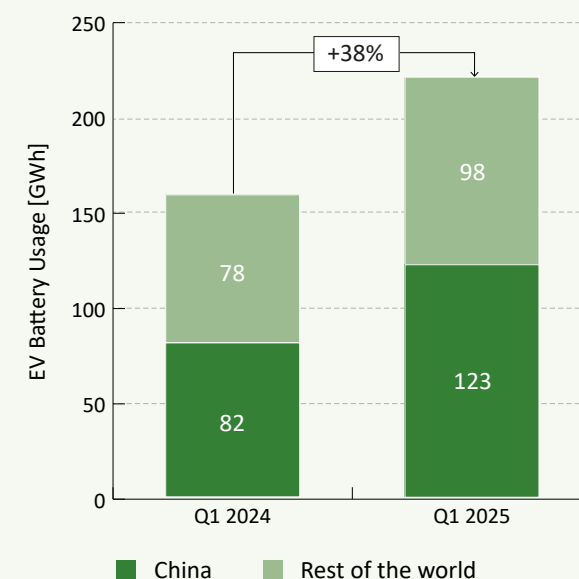


on the electric vehicle market's trajectory and forecasts global sales growth of 25%, surpassing 20 million vehicles sold in 2025.

Global EV and ESS battery market grows in 2024

The positive development of the global EV market is having a corresponding impact on the lithium-ion battery market. According to [SNE Research](#), a total of 998 GWh of lithium-ion batteries for EVs were delivered worldwide in 2024, a growth of around 15% year-on-year. The global market for stationary energy storage systems (ESS) also grew in addition to the demand for traction batteries for EVs. Last

Figure 4: Comparison of the global lithium-ion battery market for EV applications in the first quarter of 2024 and 2025.



year, 301 GWh of lithium-ion batteries were delivered for ESS applications. This corresponds to growth of over 60% compared to the previous year. Overall, the EV and ESS market generated a demand for lithium-ion batteries of approximately 1,300 GWh in 2024.

This positive trend continued in the first quarter of 2025. According to [SNE Research](#), approximately 220 GWh of lithium-ion batteries were installed in delivered electric vehicles globally, representing an increase of around 60 GWh (an increase of roughly 40%) compared to the same quarter of the previous year. With an increase of

40 GWh, the greatest growth was achieved in China. [Outside of China](#), deliveries increased by approximately 20 GWh compared to the same quarter of the previous year. By the end of the year, sales of traction batteries in the automotive sector alone could reach around 1,200 GWh.

Lithium-ion battery cell production in Europe

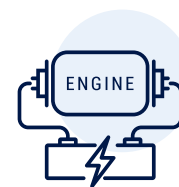
The stagnation of the European EV market last year impacted the development of new cell production capacities as well as the expansion of the upstream and downstream stages of the value chain in Europe. Compared to the [Market Update Q2 2024](#), only one company, SK On, has commissioned a new cell plant and started series production. The [production commenced](#) in the second quarter of 2024. The new plant in Iváncsa, Hungary, has a nominal production capacity of [30 GWh/a](#), which is expected to be fully operational in 2025. Morrow also opened a battery factory in Norway last year. Due to [delays](#), however, production cannot start until 2025. The factory will produce [LFP battery cells](#) for stationary energy storage systems, among other things.

In addition to the openings, there were also setbacks for the European lithium-ion battery industry last year. In November, [Northvolt](#) initiated a Chapter 11 bankruptcy proceeding in the USA to restructure its business. Despite this proceeding and other efforts, the financial conditions for the continuation of the company could not be established, leading Northvolt to file for [bankruptcy](#) in Sweden in March 2025. Due to the continuation of limited operations at least until [the end of June 2025](#), Northvolt is listed under 'In operation' with its nominal production capacity in Figure 5.

Overall, the nominal production capacity of lithium-ion battery cells in Europe totals approximately 210 GWh/a.

However, the example of Northvolt illustrates that nominal production capacity is a theoretical value that does not equate to actual production. The ramp-up of production is particularly challenging and can lead to significant discrepancies between production capacity and actual output. This especially applies to inexperienced cell manufacturers, which are highlighted by an 'X' in the last column under the 'In operation' category in Figure 5.

Compared to the Market Update Q2 2024, three new projects have been added to the 'Under construction' category: [AESC](#) in Spain, [AGRATAS](#) in the UK and [CALB](#) in Portugal. All of these projects held their groundbreaking ceremonies last year. In addition to SK On and Morrow, which have commenced operations or opened their plants, NOVO Energy is no longer listed in the 'Under construction' category. The former joint venture between Northvolt and [Volvo Cars](#) was fully acquired by Volvo Cars at the beginning of this year. Volvo Cars is now looking for a new technology partner with the necessary cell production expertise. The goal of establishing long-term cell production remains, but due to the economic situation, NOVO Energy is now pursuing these efforts to a limited extent. For this reason, NOVO Energy's project in Figure 5 has been moved to the new 'On hold' category. Cellforce and PowerCo in Germany, AESC and Verkor in France, and CATL in Hungary listed in the 'Under construction' category are aiming to start production this year. Additionally, [ACC](#) plans to commission the second block in Billy-Berclau, France, and AESC intends to launch its second [gigafactory in Sunderland](#), UK, and prepare for mass production. These projects could increase the nominal production capacity in Europe by more than 100 GWh/a over the next one to two years.



Since the Market Update Q2 2024, a new joint venture between [CATL and Stellantis](#) in Spain has been announced. The companies are planning cell production for LFP batteries with an annual capacity of up to 50 GWh. Inobat's project in Slovakia is now listed as the joint venture [Gotion Inobat Batteries](#) (GIB). The announced projects by ACC in Germany and Italy are currently on hold and are therefore listed in the corresponding category. The company [SVolt](#), which has discontinued its operating business in Europe, is no longer listed.

In addition to the companies in Figure 5 which focus on producing lithium-ion cells for traction batteries, there are further manufacturers specialising in lithium-ion cells for special applications. These companies are likewise confronted with a challenging market environment.

European policies to strengthen the battery industry: Clean Industrial Deal, Automotive Action Plan and Strategic Projects

With the new EU Commission taking office, a strategic adjustment of European industrial policy is becoming apparent. In addition to the consistent pursuit of climate neutrality objectives, the focus is shifting to strengthening Europe's competitiveness and geopolitical resilience. The [Clean Industrial Deal](#) (CID) adopted in February 2025 provides a central framework for this.

The aim of the CID is to reinforce the long-term viability and competitiveness of European industry without abandoning the EU's ambitious climate neutrality targets. Ensuring secure and affordable energy, targeted investment in innovative and sustainable technologies and access to raw materials is key to achieving these objectives.

Figure 5: Production of lithium-ion battery cells in Europe. The focus is on the production of cells for traction batteries.

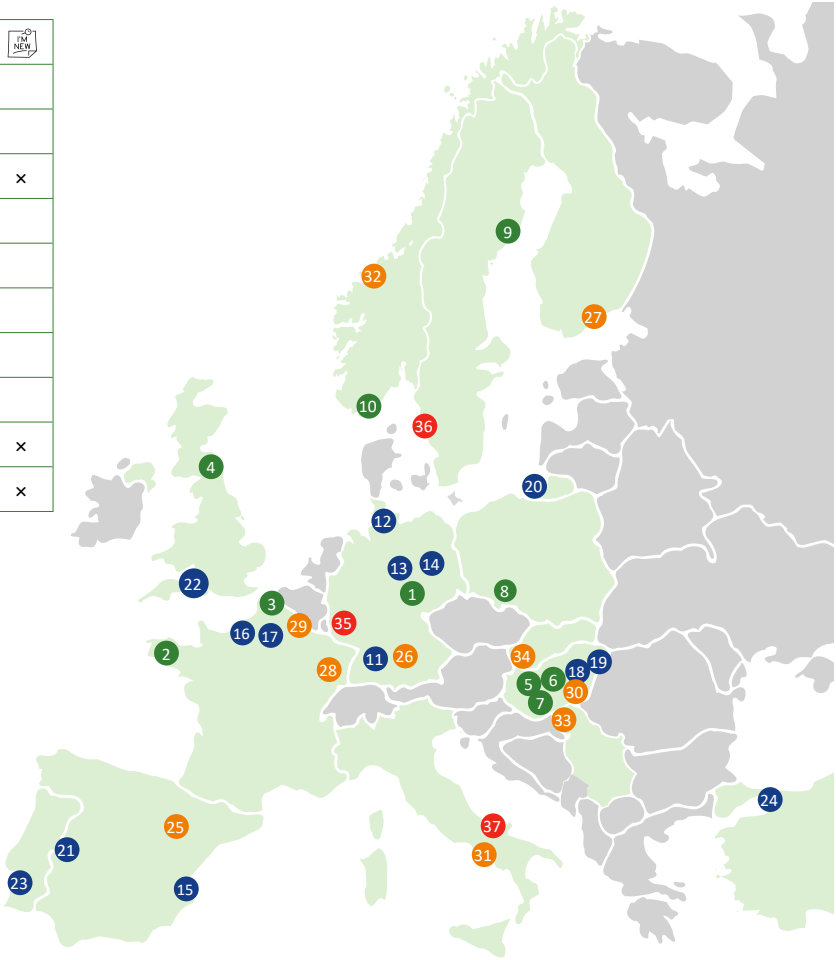
In operation

#	Company						
1	CATL	DE	2022	14	24	24	
2	BlueSolutions	FR	2013	1	1	1	
3	ACC	FR	2023	15	28	40	×
4	AESC	GB	2012	2	14	35	
5	SK on	HU	2020	18	18	18	
6	SAMSUNG SDI	HU	2018	40	50	50	
7	SK on	HU	2024	20	30	30	
8	LG Energy Solution	PL	2018	86	90	115	
9*	northvolt	SE	2021	16		60	×
10	MORVOW	NO	2024	1		43	×

Under construction

#	Company						
11	CELLFORCE	DE	2025		1	1	
12	northvolt	DE				60	
13	PowerCo	DE	2025		20	40	
14	Tesla	DE				100	
15	PowerCo	ES	2026		40	60	
16	AESC	FR	2025		9	30	
17	Vekor	FR	2025		16	50	
18	CATL	HU	2025		40	100	
19	EVE 亿纬锂能	HU	2026		28	28	
20	Rosatom	RU	2026		4	12	

*Operation will be discontinued on 30/06/2025




Start of
Production


Capacity [GWh/a]
In operation | Build-up to
(Planning first phase) |
Maximum capacity


Newcomers vs. Experts:
While experienced manufacturers can
complete a ramp-up within several months,
newcomers may require over a year.

#	Company					
21	AESC	ES	2026			30
22	AGRATAS	GB	2026			40
23	CALB	PT	2028		15	45
24	siro	TR	2026		20	50

Announced

#	Company					
25	STELLANTIS CATL	ES	2026			50
26	VARTA	DE				2
27	FINNISH MINERALS GROUP	FI				60
28	BlueSolutions	FR	2030			25
29	ProLogium	FR	2026			48
30	SUNJODA	HU				
31	FAAM	IT	2026		8	8
32	e linor.	NO	2026			40
33	ElevenEs	RS	2026		8	48
34	GIB	SK	2026		20	40

On hold

#	Company					
35	ACC	DE			13	40
36	NOVO	SE			18	50
37	ACC	IT			26	40

 Investment: around
EUR 75–120 million per GWh  Jobs: around 50–100 jobs
per GWh

Source: Public announcements, own depiction.

The CID explicitly addresses batteries and energy storage as a key technology. State aid rules are to be simplified to give member states the opportunity to provide greater support for strategic clean-tech industries such as batteries. To this end, a new [Clean Industrial State Aid Framework \(CISAF\)](#) is to be established, which will replace the existing Temporary Crisis and Transition Framework (TCTF).

The EU Commission plans to accelerate the expansion of battery storage systems through simplified approval procedures and to support the industrialisation of innovative battery technologies, also through the European Investment Bank (EIB). The integration of sustainability criteria into public and private procurement processes is intended to stimulate the demand for green batteries, preferably manufactured in Europe.

In order to secure the supply of the European battery recycling industry, black mass was categorised as hazardous waste and exports to non-OECD countries are therefore prohibited.

Another key component is the establishment of the *Transregional Circularity Hubs*, which are intended to promote targeted cooperation between member states and industry players for the recycling of essential materials. A further pillar of the CID is the protection of European industry from international overcapacity through customs duties.

Prior to the publication of the CID, associations such as [RECHARGE](#) recommended various measures and their rapid implementation. In particular, upscaling should be strengthened through production support. The demand

for European batteries (and EVs) should be increased and a level playing field for the European battery industry should be established. At the same time, bureaucratic hurdles should be reduced, and regulations harmonised across Europe.

Automotive Action Plan to strengthen the automotive industry

Building on the CID, the [Automotive Action Plan \(AAP\)](#) was introduced as an industrial policy signal for transforming and strengthening the competitiveness of the European automotive sector. A central part of the AAP is the establishment of a competitive and innovative European battery industry, with the objective of achieving an EU-internal value-added share of over 50% by 2030.

This includes not only cell manufacturing but also upstream stages such as the production of active materials and the extraction and processing of raw materials – both within the EU and through strategic partnerships abroad. To finance this ambition, the EU

Commission plans to utilise a range of instruments. In particular, a *battery booster* is planned to achieve the competitiveness of batteries manufactured in Europe.

- Overall, up to EUR 3 billion are to be made available through the Innovation Fund for the manufacturing of EV battery cells.
- The facilitations for state aid already mentioned in the CID are intended to simplify state support for investment projects. In this context, the EU Commission is also analysing possible aid for the upscaling of production lines and direct production support in combination with state aid.

- The InvestEU fund will provide an additional EUR 200 million in the form of venture debt funding in the years 2025–2027 to finance innovative projects along the European battery value chain.
- To promote research on next-generation batteries, the EU Commission is allocating approximately EUR 350 million from the Horizon Europe programme and the Batt4EU partnership for the period 2025–2027.

Furthermore, future legal requirements for the European content of battery cells and components in EVs are to be introduced (European Content Requirements). In order to achieve a level playing field, the EU Commission intends to assess foreign direct investments based on their contribution to European value creation and, where necessary, introduce measures, such as rules for joint ventures. Information on the repair, maintenance and condition of batteries is to be made available from Q3 2026 by means of the digital battery passport.

Other planned measures to boost the BEV ramp-up include the expansion of the charging infrastructure, especially for electric commercial vehicles, and social leasing models.

In addition to the measures in the CID, the establishment of a Critical Raw Materials Centre by 2026 is intended to increase transparency on the raw materials markets, pool demand and coordinate joint investments to secure the supply of raw materials. The strategic projects under the Critical Raw Material Act (CRMA) serve as a further measure.

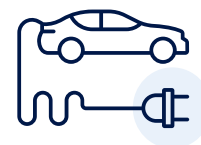
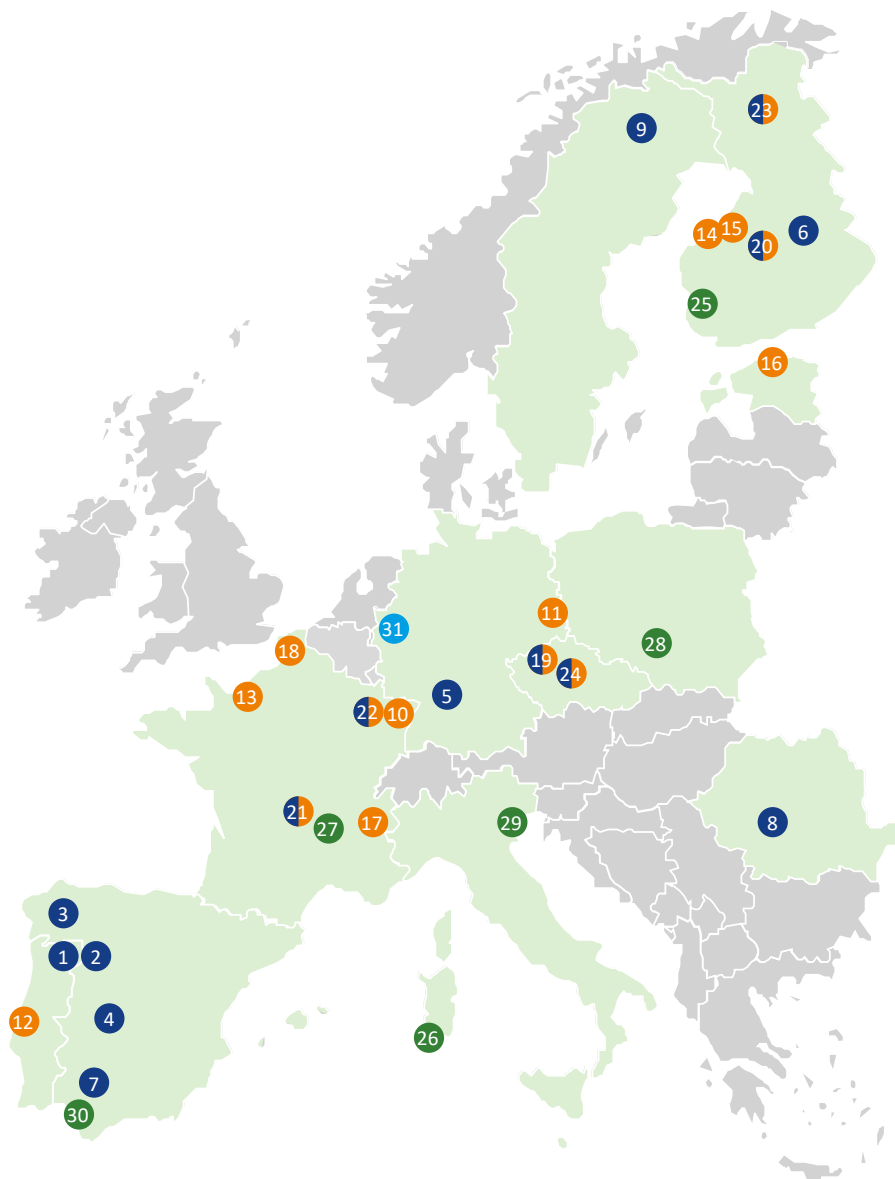


Figure 6: Strategic projects approved under the Critical Raw Materials Act relating to the battery raw materials lithium, nickel, manganese, cobalt, graphite and silicon.

Extraction			
1	LUS • RECURS • S	Li	
2	SAVANNAH	Li	
3	RDC REGIONES DE GALICIA	Li	
4		Li	
5	VULCAN ENERGY	Li	
6	Terrafame	Ni Co	
7	DENARIUS METALS	Ni Co	
8		C	
9	talga	C	

Processing			
10	VIRIDIAN LITHIUM	Li	
11	RockTech Lithium	Li	
12	Lifthium Elevating energy	Li	
13	Sibanye Stillwater	Li Ni Mn Co	
14	Jervois FINLAND	Co	
15	hycamite	C	
16	UP CATALYST	C	
17	TOKAI COBEX	C	
18	NORTHERN GRAPHITE	C	



Extraction and Processing			
19		Li	
20	Sibanye Stillwater	Li	
21	IMERYS	Li	
22	eramET	Li	
23	AngloAmerican	Ni Co	
24	EURO MANGANESE	Mn	

Recycling			
25	fortum	Li Ni Co	
26		Li Ni Co	
27	orano	Li Ni Mn Co	
28	elemental strategic materials	Li Ni Mn Co	
29	CIRCULAR MATERIALS	Ni	
30	ATLANTIC COPPER	Ni	

Substitution			
31	pcc Thorion	C Si	

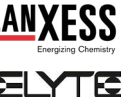
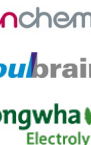
Strategic projects for raw material extraction and processing

As part of the CRMA, the EU has established a procedure for the selection of so-called [Strategic Projects](#). These make a special contribution to the supply of 14 critical

raw materials that are essential for industry in Europe – especially for key technologies such as batteries. In future, there will be annual calls for funding to select new projects.

Strategic Projects cover the stages of raw material extraction, processing and recycling and must contribute to a secure, economically viable, sustainable, and diversified European raw material supply. They benefit from accelerated approval procedures (maximum 18 months), funding priority at both EU and national levels, and easier access to financing instruments such as InvestEU or the European Investment Bank.

Figure 7: Companies that produced lithium-ion battery cells or components for traction batteries in Europe in 2024. Locations that have not yet started production in 2024 are not included.

	Nameplate capacity Europe 2024e	 Sweden	 United Kingdom	 Germany	 France	 Poland	 Hungary
Battery cell 	 210 GWh/a	 					
Cathode-activematerial 	 50 GWh _{eq} /a			 IBU tec			
Separator 	 170+ GWh _{eq} /a						
Electrolyte 	 400 GWh _{eq} /a						

 Filed for bankruptcy

Elektrolyte: 1 kt ≈ 1 GWh_{eq}; Separator: 1 Mio. m² ≈ 0,1 GWh_{eq}

References: Company announcements, own depiction

The first [47 projects](#) were selected in March 2025. Of these, 31 projects focus on raw materials that are central to battery production: Lithium, cobalt, nickel, manganese and graphite.

The upstream value chain stage of cell production: Component production in Europe

Figure 7 provides a concise overview of cell and component manufacturing in Europe. This figure only includes plants that have been producing in Europe up until the end of 2024. Planned facilities or those currently under construction are not included. This matrix was already used in the [Market Update Q2 2023](#) to illustrate active cell and component production in 2023. With the start of production at the end of 2023, ACC complements the row of cell manufacturers. As previously mentioned, the nominal production capacity of the operational cell manufacturing plants is approximately 210 GWh/a. In the field of cathode active material manufacturing, Ibu-tec is newly listed. The production capacities of [Umicore](#) in Poland is now around 30 GWh_{eq}/a, resulting in a total nominal production capacity of around 50 GWh_{eq}/a. Regarding separator production, Europe has nominal production capacities of more than 170 GWh_{eq}/a. No companies have been added since the Market Update Q2 2023. The capacity for electrolyte production has now reached approximately 400 GWh_{eq}/a. A large part of

the production capacity is attributed to [Enchem](#), which according to its own statements had a capacity of 105 kt/a in Poland and in Hungary, respectively. A new company, [E-Lyte](#), has been added to the list in Germany.

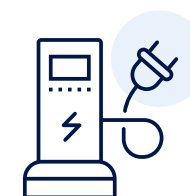
While the electrolyte capacity is more than sufficient to supply cell production, there are still discrepancies with the remaining components. The manufacturing capacity for separators is approaching that of cell production. Since capacity data is not available for all separator manufacturers listed in the matrix, the actual capacity exceeds the stated 170 GWh_{eq}/a. Additionally, new plants and expansions are under construction, including those of [W-Scope](#) in Hungary, [SK IE Technologies](#) in Poland and [Senior Material](#) in Sweden.

Significant discrepancies persist between the capacity for cathode active materials (CAM) and cell manufacturing capacity, meaning Europe remains heavily reliant on imports in this area. In the next one to two years, manufacturing capacities are expected to increase, as four CAM plants are currently being built in Europe. [Ecopro](#) plans to commission the new plant in Debrecen, Hungary, in the first quarter of 2026. Furthermore, Huayou Cobalt is currently building a new CAM factory in Hungary, Ionway (JV of Umicore and PowerCo) in Poland and [Easpring Finland](#) (JV of Finish Minerals Group and Beijing Easpring Materials) in Finland. A detailed overview of CAM production in Europe is provided in the [Market Update Q4 2024](#).

Production of anode active materials (AAM) continues to face a substantial shortfall. As shown in the [Market Update Q2 2024](#), there are currently only a few and rather small production facilities in Europe. The construction

and expansion of new and existing production facilities is planned although it is not without complications. For example, [Putailai](#) has cancelled its plans to build an AAM factory with a capacity of up to 100 kt/a in Sweden. One of the reasons for this is the high regulatory requirements resulting from a foreign direct investment law that came into force in Sweden in 2024. [Tokai Cobex](#) has temporarily halted further investments in expanding AAM production in France due to the challenging market environment.

Despite the continuing large deficit in AAM production, the overall situation has improved in the first two quarters of 2025 compared to the difficult year 2024. This is primarily driven by rising EV sales and the resulting increase in demand for traction batteries. A growing EV market is one of the prerequisites for the establishment and development a lithium-ion battery industry in Europe. Despite some setbacks, production capacities in Europe are being continuously established and expanded to meet the rising demand. This applies not only to cell production, but also to the components directly required for it. The EU has announced and initiated policies to support the still young lithium-ion battery industry encompassing both upstream and downstream segments of the value chain. These measures could prove decisive in strengthening European companies within the competitive lithium-ion battery market and enhancing Europe's resilience.



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